# Model 5 (v3): Estimation Without Observing Public Covariates

This document addresses the identification and estimation of the structural model developed in v2 (`M5_convert4intostructural_v2.tex`) when the econometrician does not observe—or only partially observes—the public covariates $\mathbf{x}$. Section 1 states the problem. Section 2 establishes a key theoretical result: the entrant's rate $r_2$ is a sufficient statistic for $\mathbf{x}$ in the stage-2 equilibrium. Section 3 develops the estimation approach. Section 4 validates the identification strategy through Monte Carlo simulation.

# 1 The Problem

## 1.1 Model Recap

The structural model (v2) features two banks competing for a borrower. Bank 1 (informed/incumbent) observes both public covariates $\mathbf{x}$ and private covariates $\mathbf{x}'$, and therefore knows the borrower's default probability $h = h(\mathbf{x}, \mathbf{x}')$; Bank 2 (uninformed/entrant) observes only $\mathbf{x}$ and faces the conditional distribution $F(h \mid \mathbf{x})$. Demand is logit:

$$s(r_1, r_2) = \frac{1}{1 + \exp\big(\alpha(r_1 - r_2) - \lambda\big)} \tag{1}$$

In equilibrium:

$$\sigma^*(h; \mathbf{x}) = \frac{1}{1-h} + \frac{1}{\alpha\big(1 - s(\sigma^*(h; \mathbf{x}), r_2^*(\mathbf{x}))\big)} \tag{2}$$

$$r_2^*(\mathbf{x}) = \frac{1}{1 - \tilde{h}(\mathbf{x})} + \frac{1}{\alpha} \cdot \frac{\int (1-s)(1-h)\, dF(\cdot|\mathbf{x})}{\int s\,(1-s)(1-h)\, dF(\cdot|\mathbf{x})} \tag{3}$$

The structural parameters are $\theta = (\alpha, \lambda, F(\cdot|\mathbf{x}))$.

## 1.2 Identification with Observable x (v2 Result)

When the econometrician observes $(r_i^{\text{win}}, W_i, D_i, \mathbf{x}_i)$, identification proceeds by conditioning on $\mathbf{x}$ (see v2, Section 7).

## 1.3 What Breaks Without x

If the econometrician does not observe $\mathbf{x}$, the identification argument breaks at multiple steps:

| Step | Why it breaks |
|---|---|
| Step 1 | $r_2^*(\mathbf{x}_i)$ varies across borrowers. Among bank 2 winners, the econometrician observes a distribution of winning rates rather than a single constant rate. Without $\mathbf{x}$, one cannot determine which rate corresponds to which "market." |
| Step 2 | Bank 1's strategy $\sigma^*(h;\mathbf{x})$ depends on $\mathbf{x}$ through $r_2^*(\mathbf{x})$. Two bank 1 winners with the same winning rate $r_1$ but different $\mathbf{x}$ have different types $h$. Without conditioning on $\mathbf{x}$: $E[D_i \mid r_1, W_i = 1] \neq h_i$ in general; it is a weighted average over the different $(h, \mathbf{x})$ pairs that produce rate $r_1$. The clean type-recovery step fails. |
| Steps 3–4 | Without clean identification of $\sigma^*(\cdot;\mathbf{x})$ and $r_2^*(\mathbf{x})$, the identifying restriction and distribution recovery cannot be applied. |

## 2 The Sufficient Statistic Result

The following result is the key to both estimation approaches developed below. It says that the entrant's rate $r_2^*(\mathbf{x})$ encodes all the information about $\mathbf{x}$ that matters for the stage-2 equilibrium behavior.

**Proposition 1** (Sufficient statistic)**.** *In the stage-2 equilibrium, the public covariates* $\mathbf{x}$ *affect bank 1's pricing rule* $\sigma^*$ *and the market shares* $s$ *only through the entrant's rate* $r_2^*(\mathbf{x})$*. Formally, for any two covariate vectors* $\mathbf{x}_a, \mathbf{x}_b \in \mathcal{X}$ *with* $r_2^*(\mathbf{x}_a) = r_2^*(\mathbf{x}_b) = r_2$*:*

$$\sigma^*(h; \mathbf{x}_a) = \sigma^*(h; \mathbf{x}_b) = \sigma^*(h; r_2), \qquad s(\sigma^*(h; \mathbf{x}_a), r_2) = s(\sigma^*(h; \mathbf{x}_b), r_2) \tag{4}$$

*for all* $h$*.*

*Proof.* Immediate from the bank 1 FOC (2):

$$\sigma^*(h; \mathbf{x}) = \frac{1}{1-h} + \frac{1}{\alpha\big(1 - s(\sigma^*(h; \mathbf{x}), r_2^*(\mathbf{x}))\big)}$$

The right side depends on $\mathbf{x}$ only through $r_2^*(\mathbf{x})$. Since bank 1's best response is the unique solution to this fixed-point equation, and the equation is identical for all $\mathbf{x}$ with the same $r_2^*$, the solution $\sigma^*(h; \mathbf{x})$ is the same. □

*Remark* 2 (Interpretation). Proposition 1 states that from bank 1's perspective—and from the consumer's perspective—all that matters about $\mathbf{x}$ is its effect on the competitor's rate $r_2^*(\mathbf{x})$. Two borrowers with different credit scores, incomes, or employment status but facing the same entrant rate $r_2^*$ experience the same equilibrium pricing from bank 1. This is economically natural: bank 1's optimal pricing trade-off between margin and retention depends on the competitive alternative $r_2^*$, not on why $r_2^*$ takes that value.

*Remark* 3 (What $r_2^*$ does not capture). The sufficient statistic result applies to bank 1's strategy and market shares, but not to the full equilibrium. Bank 2's FOC (3) depends on $F(h|\mathbf{x})$, which is not a function of $r_2^*$ alone. Two different $\mathbf{x}$ values can lead to different $F(h|\mathbf{x})$ but the same optimal $r_2^*$.[1] However, for the econometrician's identification problem, we show below that conditioning on $r_2$ is sufficient.

**A key consequence:** The sufficient statistic result implies that the econometrician can replace $\mathbf{x}$ with $r_2$ as the conditioning variable throughout the identification argument. This motivates the two approaches below, organized by what data is available.

[1] This is analogous to different cost distributions generating the same equilibrium price in a standard pricing model.

# 3 Estimation

## 3.1 Data

For each borrower $i$, the econometrician observes:

| Variable | Description | Values |
|---|---|---|
| $r_i^{\text{win}}$ | Interest rate charged by the chosen bank | $\mathbb{R}_+$ |
| $W_i$ | Identity of the chosen bank | $\{1, 2\}$ |
| $D_i$ | Default indicator | $\{0, 1\}$ |
| $\mathbf{z}_i$ | Observed public covariates (possibly empty) | $\mathbb{R}^{d_z}$ |

The econometrician may observe all, some, or none of the public covariates $\mathbf{x}$. We partition $\mathbf{x} = (\mathbf{z}, \mathbf{w})$, where $\mathbf{z}$ is observed and $\mathbf{w}$ is not. The special case $\mathbf{z} = \emptyset$ (no covariates observed) is included.

**Key asymmetry:**

- When $W_i = 2$: we observe $r_i^{\text{win}} = r_2^*(\mathbf{x}_i)$, so the entrant's rate is directly revealed.
- When $W_i = 1$: we observe $r_i^{\text{win}} = \sigma^*(h_i; r_2^*(\mathbf{x}_i))$, which depends on the borrower's type $h_i$ <u>and</u> the latent entrant rate $r_2^*(\mathbf{x}_i)$.

## 3.2 Notation and the Role of the Sufficient Statistic

By Proposition 1, bank 1's equilibrium pricing strategy $\sigma^*(h; \mathbf{x})$ depends on $\mathbf{x}$ only through $r_2^*(\mathbf{x})$, so we write $\sigma^*(h; r_2)$ without ambiguity. This has a critical implication: the mapping from the borrower's type $h$ to the incumbent's rate $r_1$ is fully determined by the scalar $r_2$.

**Definition 4** (Type-recovery function)**.** For fixed $r_2$, define $\tau(r_1; r_2)$ as the inverse of $\sigma^*(\cdot\,; r_2)$ with respect to $h$:

$$\tau\big(\sigma^*(h; r_2);\, r_2\big) = h \qquad \text{for all } h \in [\underline{h}(r_2),\, \bar{h}(r_2)] \tag{5}$$

Since $\sigma^*(\cdot; r_2)$ is strictly increasing in $h$ (higher default probability $\Rightarrow$ higher rate), $\tau(\cdot; r_2)$ is well-defined and strictly increasing in $r_1$.

**Interpretation.** Given an observed bank 1 winning rate $r_1$ and a hypothesized entrant rate $r_2$, the function $\tau(r_1; r_2)$ returns the unique default probability $h$ that would generate rate $r_1$ in equilibrium. Without the sufficient statistic result, this inverse would also depend on the full vector $\mathbf{x}$, making the likelihood intractable.

The Jacobian of the transformation is:

$$\tau'(r_1; r_2) \equiv \frac{\partial \tau}{\partial r_1}(r_1; r_2) = \frac{1}{\sigma_h^*\big(\tau(r_1; r_2);\, r_2\big)} > 0 \tag{6}$$

which is needed for the change-of-variables from $h$ to $r_1$ in the likelihood.

## 3.3 Information from Bank 2 Winners

Among bank 2 winners, $r_i^{\text{win}} = r_2^*(\mathbf{x}_i)$ is directly observed. This reveals two objects:

1. **Distribution of entrant rates (under selection).** The empirical distribution of $r_i^{\text{win}}$ among bank 2 winners with covariates $\mathbf{z}$ identifies:

$$g_{r_2|W=2,\mathbf{z}}(r_2) = \frac{g(r_2|\mathbf{z}) \cdot \Pr(W = 2 \mid r_2, \mathbf{z})}{\Pr(W = 2 \mid \mathbf{z})} \tag{7}$$

   where $g(r_2|\mathbf{z})$ is the density of $r_2^*(\mathbf{x})$ conditional on $\mathbf{z}$ and $\Pr(W = 2 \mid r_2, \mathbf{z}) = \int (1 - s(\sigma^*(h; r_2), r_2))\, dF(h|r_2, \mathbf{z})$ is the aggregate switching probability.

   This is informative about $g(\cdot|\mathbf{z})$ but confounded by selection: borrowers facing lower $r_2$ (more competitive entrants) are more likely to switch, so the observed distribution among bank 2 winners is shifted downward relative to the population distribution.

2. **Default rates conditional on $r_2$.** For each value of $r_2$ (and $\mathbf{z}$):

$$E[D_i \mid r_i^{\text{win}} = r_2, W_i = 2, \mathbf{z}] = \frac{\int h\,(1 - s(\sigma^*(h; r_2), r_2))\, dF(h|r_2, \mathbf{z})}{\int (1 - s(\sigma^*(h; r_2), r_2))\, dF(h|r_2, \mathbf{z})} \tag{8}$$

   This is the adverse-selection-weighted default rate. It is <u>not</u> the unconditional mean $E[h \mid r_2, \mathbf{z}]$ because the switching probability $1 - s$ varies with $h$: borrowers with lower $h$ (good types) face a lower $r_1 = \sigma^*(h; r_2)$, closer to the entrant's $r_2$, making them more price-sensitive and more likely to switch. This creates adverse selection <u>for the incumbent</u>: the borrowers who leave are disproportionately good.

## 3.4 The Challenge for Bank 1 Winners

Among bank 1 winners, the observed rate $r_1 = \sigma^*(h_i; r_2^*(\mathbf{x}_i))$ depends on two unobservables: the type $h_i$ and the entrant rate $r_2^*(\mathbf{x}_i)$. Even conditioning on $\mathbf{z}$, the entrant rate $r_2$ varies through the unobserved $\mathbf{w}$.

**The fundamental identification problem.** The same winning rate $r_1$ can arise from different $(h, r_2)$ combinations:

- A borrower with low $h$ (good type) facing a low $r_2$ (competitive entrant): bank 1 offers a low rate to retain this good borrower against strong competition.
- A borrower with higher $h$ (riskier type) facing a higher $r_2$ (less competitive entrant): bank 1 charges more because both the cost of lending and the competitive slack are higher.

Both scenarios can produce the same observed $r_1$. The expected default rate conditional on observing $r_1$ among bank 1 winners is therefore a <u>mixture</u>:

$$E[D_i \mid r_i^{\text{win}} = r_1, W_i = 1, \mathbf{z}] = \frac{\int_{\mathcal{R}_2(r_1)} \tau(r_1; r_2) \cdot s(r_1, r_2) \cdot f(\tau(r_1; r_2)|r_2, \mathbf{z}) \cdot \tau'(r_1; r_2) \cdot g(r_2|\mathbf{z})\, dr_2}{\int_{\mathcal{R}_2(r_1)} s(r_1, r_2) \cdot f(\tau(r_1; r_2)|r_2, \mathbf{z}) \cdot \tau'(r_1; r_2) \cdot g(r_2|\mathbf{z})\, dr_2} \tag{9}$$

where $\mathcal{R}_2(r_1) = \{r_2 : \tau(r_1; r_2) \in [\underline{h}(r_2), \bar{h}(r_2)]\}$. This weighted average over $r_2$ does <u>not</u> equal $h_i$ for any individual borrower, so the clean type-recovery step from v2 fails.

**The role of z.** Conditioning on observed covariates $\mathbf{z}$ reduces the residual variation in $r_2$ (since $r_2$ varies only through $\mathbf{w}$ conditional on $\mathbf{z}$), which tightens the mixture. In the limit where $\mathbf{z} = \mathbf{x}$ (all public covariates observed), $r_2^*(\mathbf{x})$ is deterministic given $\mathbf{z}$, the mixture collapses, and we recover the nonparametric identification of v2. However, nonparametric identification fails whenever any component of $\mathbf{x}$ is unobserved, necessitating parametric structure.

## 3.5 Parametric Specification

Since nonparametric identification fails without fully observing $r_2$ for bank 1 winners, we impose parametric structure on two objects.

*Assumption* 5 (Parametric specification). (i) **Type distribution.** $h \mid (r_2, \mathbf{z})$ follows a parametric family $F(h; \theta_F(r_2, \mathbf{z}))$, where $\theta_F(r_2, \mathbf{z})$ collects the distribution parameters. For example, $h \mid (r_2, \mathbf{z}) \sim \text{Beta}(a(r_2, \mathbf{z}), b(r_2, \mathbf{z}))$, where $a$ and $b$ are functions of $(r_2, \mathbf{z})$ specified up to finite-dimensional parameters.[2]

(ii) **Mixing distribution.** $r_2 \mid \mathbf{z} \sim G(\cdot \mid \mathbf{z}; \phi)$ belongs to a parametric family.

When no covariates $\mathbf{z}$ are observed, the conditioning on $\mathbf{z}$ is dropped: $F(h; \theta_F(r_2))$ and $G(r_2; \phi)$.

*Remark* 6 (Nature of the two parametric choices). Parts (i) and (ii) play different roles:

- Part (i) is the substantive economic restriction. It constrains how the distribution of borrower types varies with the competitive environment $r_2$ and observable characteristics $\mathbf{z}$. This determines the relationship between rates, defaults, and market shares that the model must fit.

- Part (ii) characterizes the population distribution of competitive conditions. While $r_2$ is directly observed among bank 2 winners, those observations are selected (borrowers who switched), so recovering the unconditional mixing density $g(r_2|\mathbf{z})$ from the observed $g(r_2|W = 2, \mathbf{z})$ requires knowing the selection probability $\Pr(W = 2|r_2, \mathbf{z})$, which depends on all the structural parameters. Consequently, $G$ cannot be estimated independently of the rest of the model, and a parametric specification is needed for the mixture integral in the bank 1 winner likelihood.

The full parameter vector is $\theta = (\alpha, \lambda, \theta_F, \phi)$.

## 3.6 Likelihood Derivation

We derive the likelihood for each observation. The logic differs by which bank wins.

[2] Flexible alternatives include the Kumaraswamy distribution on $[\underline{h}(r_2), \bar{h}(r_2)]$ or a beta regression specification.

### 3.6.1 Bank 1 wins ($W_i = 1$): observed $r_i^{\mathbf{win}} = \sigma^*(h_i; r_2)$, latent $r_2$

Consider a borrower with type $h$ and entrant rate $r_2$, with observed covariates $\mathbf{z}$. The "complete data" density for observing $(r_1, r_2, W{=}1, D)$ factorizes as:

$$p(r_1, r_2, W{=}1, D \mid \mathbf{z}, \theta) = \underbrace{f\big(\tau(r_1; r_2) \mid r_2, \mathbf{z}; \theta_F\big) \cdot \tau'(r_1; r_2)}_{\substack{\text{density of type } h=\tau(r_1;r_2) \\ \text{transformed to the rate scale}}} \cdot \underbrace{s(r_1, r_2)}_{\substack{\text{prob. bank 1} \\ \text{wins}}} \cdot \underbrace{h^D(1{-}h)^{1-D}}_{\text{default outcome}} \cdot \underbrace{g(r_2 \mid \mathbf{z}; \phi)}_{\substack{\text{density of} \\ \text{entrant rate}}} \tag{10}$$

where $h = \tau(r_1; r_2)$ throughout. The first term applies the change-of-variables formula: $h$ has density $f(h|r_2, \mathbf{z})$; transforming to $r_1 = \sigma^*(h; r_2)$ yields density $f(\tau(r_1; r_2)|r_2, \mathbf{z}) \cdot \tau'(r_1; r_2)$.

Since $r_2$ is **unobserved** for bank 1 winners, we marginalize:

$$\boxed{\mathcal{L}_i^{(1)}(\theta) = \int_{\mathcal{R}_2(r_i^{\text{win}})} f\big(\tau(r_i^{\text{win}}; r_2) \mid r_2, \mathbf{z}_i; \theta_F\big) \cdot \tau'(r_i^{\text{win}}; r_2) \cdot s(r_i^{\text{win}}, r_2) \cdot h^{D_i}(1{-}h)^{1-D_i} \cdot g(r_2 \mid \mathbf{z}_i; \phi)\, dr_2} \tag{11}$$

with $h = \tau(r_i^{\text{win}}; r_2)$ inside the integrand. The integration domain $\mathcal{R}_2(r_i^{\text{win}})$ is the set of entrant rates $r_2$ for which the implied type $\tau(r_i^{\text{win}}; r_2)$ lies in the support of $F(\cdot|r_2, \mathbf{z}_i)$.

### 3.6.2 Bank 2 wins ($W_i = 2$): observed $r_i^{\mathbf{win}} = r_2$, latent $h$

When bank 2 wins, the entrant rate $r_i^{\text{win}} = r_2$ is directly observed, but the borrower's type $h$ is latent—we only see the default outcome $D_i$. Integrating over all types that could have switched:

$$\boxed{\mathcal{L}_i^{(2)}(\theta) = g(r_i^{\text{win}} \mid \mathbf{z}_i; \phi) \cdot \int_{\underline{h}(r_i^{\text{win}})}^{\bar{h}(r_i^{\text{win}})} \big(1 - s(\sigma^*(h; r_i^{\text{win}}), r_i^{\text{win}})\big) \cdot h^{D_i}(1-h)^{1-D_i} \cdot f(h \mid r_i^{\text{win}}, \mathbf{z}_i; \theta_F)\, dh} \tag{12}$$

The first factor $g(r_i^{\text{win}} \mid \mathbf{z}_i; \phi)$ is the density of drawing a borrower with entrant rate $r_i^{\text{win}}$ from the subpopulation with covariates $\mathbf{z}_i$. The integral averages over all types $h$ consistent with that entrant rate, weighted by: (i) the probability that type $h$ switches to bank 2, i.e., $1 - s$; (ii) the default likelihood $h^{D_i}(1-h)^{1-D_i}$; and (iii) the type density $f(h|r_2, \mathbf{z}_i)$.

### 3.6.3 Full log-likelihood

$$\ell(\theta) = \sum_{i=1}^{N} \Big[\mathbf{1}(W_i = 1) \ln \mathcal{L}_i^{(1)}(\theta) + \mathbf{1}(W_i = 2) \ln \mathcal{L}_i^{(2)}(\theta)\Big] \tag{13}$$

## 3.7 Identification

Although fully nonparametric identification fails, the parametric model is identified from several independent data features:

| Parameter(s) | Identifying variation |
|---|---|
| $\phi$ (distribution of $r_2$) | The distribution of winning rates among bank 2 winners. This is the most directly informative data feature, though it is confounded by selection into bank 2. |
| $\theta_F$ (type distribution) | Default rates among bank 2 winners at different $r_2$ levels, which reveal $E[h\|r_2, \mathbf{z}, W = 2]$—the adverse-selection-weighted mean type. Combined with the distribution of rates and defaults among bank 1 winners, which depends on $F$ through the mixture (11). |
| $\alpha$ (rate sensitivity) | The elasticity of bank 2's market share with respect to rate differentials. When $\alpha$ is large, small rate differences strongly affect switching, compressing the distribution of winning rates and increasing the share of bank 2 winners. |
| $\lambda$ (switching cost) | The overall share of bank 1 winners (baseline retention rate). A higher $\lambda$ shifts demand toward the incumbent, increasing $\Pr(W=1)$ at all rate levels. |

*Remark* 7 (Role of default data). Default outcomes $D_i$ play a crucial identifying role. Without defaults, the econometrician observes only rates, winner identity, and market shares, which creates a severe identification problem: many $(\alpha, \lambda, F, G)$ combinations can generate similar rate distributions. The default indicator provides a noisy signal of the type $h$, which disciplines the mixture decomposition. In particular, high default rates among bank 1 winners with high rates confirms that those rates correspond to risky types (and not merely to weak competition from the entrant).

*Remark* 8 (Identification power from bank 2 winners). Bank 2 winners provide disproportionate identifying power: they directly reveal $r_2$, pinning down $g(\cdot|\mathbf{z};\phi)$ and the conditional relationship between $r_2$ and defaults. The more bank 2 winners in the sample, the better the identification of the mixing distribution $g$ and hence the better the disentangling of $(h, r_2)$ for bank 1 winners.

## 3.8 The Role of Observed Covariates

The richer $\mathbf{z}$ is, the better the estimation performs, through three channels:

1. **Cross-z variation.** The structural parameters $(\alpha, \lambda)$ are common across all values of $\mathbf{z}$, while $F(h|r_2, \mathbf{z})$ and $G(r_2|\mathbf{z})$ vary. Each $\mathbf{z}$-stratum generates an independent set of moments for $(\alpha, \lambda)$, providing overidentification.

2. **Tighter mixture.** Conditioning on $\mathbf{z}$ reduces the residual variance of the latent $r_2$, since $r_2^*(\mathbf{z}, \mathbf{w})$ varies only through $\mathbf{w}$ conditional on $\mathbf{z}$. This concentrates the integral in (11) on a narrower range, improving the mapping from observed rate $r_1$ to type $h$.

3. **Nonparametric limit.** When $\mathbf{z} = \mathbf{x}$ (all public covariates observed), $r_2^*(\mathbf{x})$ is deterministic given $\mathbf{z}$, the mixing distribution $G(r_2|\mathbf{z})$ degenerates to a point mass, the integral in (11)

collapses, and estimation reduces to the nonparametric v2 identification. As $\mathbf{z}$ becomes richer, the approach smoothly interpolates toward this nonparametric limit.

*Remark* 9 (No covariates: $\mathbf{z} = \emptyset$). When no covariates are observed, the conditioning on $\mathbf{z}$ is dropped everywhere: $f(h|r_2, \mathbf{z})$ becomes $f(h|r_2)$ and $g(r_2|\mathbf{z})$ becomes $g(r_2)$. The likelihood equations (11)–(12) remain valid with this simplification. This is the most data-constrained case: the mixture integral for bank 1 winners spans the full range of $r_2$ in the population, and identification relies entirely on the parametric assumptions for $F$ and $G$.

*Remark* 10 (Practical data sources for $\mathbf{z}$). In Chilean administrative credit data (CMF), likely candidates for $\mathbf{z}$ include:

- Credit bureau score (publicly available to all lenders)
- Loan amount and term (observed in regulatory filings)
- Geographic region or bank branch
- Year/month of origination (controls for time-varying market conditions)

Internal bank variables such as checking account balance, payment history on existing products, and detailed employer verification are examples of $\mathbf{w}$: observed by both banks but typically not available to the econometrician.

## 3.9 Computational Strategy

The likelihood for bank 1 winners requires a one-dimensional numerical integral over $r_2$ at each observation and each parameter evaluation. Combined with the inner equilibrium solve $\sigma^*(h; r_2)$ and the inversion $\tau(r_1; r_2)$, this is computationally expensive. A practical strategy:

1. Pre-compute $\sigma^*(h; r_2)$ on a fine grid of $(h, r_2)$ values.
2. Construct $\tau(r_1; r_2)$ by numerical inversion on the same grid.
3. Evaluate the integral in (11) by quadrature over the pre-computed grid.
4. Re-compute the grid only when $(\alpha, \lambda)$ change; the type distribution parameters $\theta_F$ do not affect $\sigma^*$ or $\tau$, so the grid can be reused across many likelihood evaluations.

# 4 Monte Carlo Simulation

To verify that the identification strategy developed in Section 3 can recover the structural parameters from finite samples, we conduct a Monte Carlo exercise. The procedure has three steps: specify the true data generating process, compute the equilibrium and generate simulated data, and check whether the identifying restrictions hold.

## 4.1 Data Generating Process

**Grids.** The public covariate $x$ (scalar for the simulation) and default probability $h$ are each discretized on equally spaced grids:

$$x \in \{x_1, \ldots, x_{N_x}\} \subset [x_{\min}, x_{\max}], \qquad h \in \{h_1, \ldots, h_{N_h}\} \subset [h_{\min}, h_{\max}]$$

**Joint distribution.** The joint distribution of $(x, h)$ is specified directly through a probability matrix $\mathbf{P} \in \mathbb{R}^{N_x \times N_h}$ with $P_{jk} = P(x = x_j,\, h = h_k) \geq 0$ and $\sum_{j,k} P_{jk} = 1$. This is the most flexible discrete specification: it imposes no parametric structure on the joint distribution and allows arbitrary dependence between $x$ and $h$. The marginal $P(x = x_j) = \sum_k P_{jk}$ and the conditional $P(h = h_k \mid x = x_j) = P_{jk}/P(x = x_j)$ are derived objects.

For the baseline simulation, $\mathbf{P}$ is constructed so that higher $x$ corresponds to riskier borrowers (stochastic dominance of $h \mid x$ shifting right as $x$ increases). Any matrix satisfying $\sum_{j,k} P_{jk} = 1$ with $P_{jk} > 0$ can be used.

**True parameters.**

| **Parameter** | **Symbol** | **Value** | **Description** |
|---|---|---|---|
| Price sensitivity | $\alpha$ | 10 | Logit demand coefficient |
| Switching cost | $\lambda$ | 0.5 | In rate units |
| Grid sizes | $N_x, N_h$ | 5, 30 | Public covariate and type grids |
| Support | $[h_{\min}, h_{\max}]$ | $[0.02, 0.40]$ | Default probability range |
| | $[x_{\min}, x_{\max}]$ | $[0.1, 0.9]$ | Public covariate range |
| Joint distribution | $\mathbf{P}$ | $N_x \times N_h$ matrix | See code for construction |
| Sample size | $N$ | 5000 | Number of simulated borrowers |

## 4.2 Equilibrium Computation

For each covariate value $x_j$, we solve for the equilibrium $(\sigma^*(\cdot; x_j), r_2^*(x_j))$ using the following nested algorithm:

1. **Initialize** $r_2$ at the break-even rate for the average type plus a markup: $r_2^{(0)} = \frac{1}{1-h(x_j)} + \frac{1}{\alpha}$.

2. **Inner solve.** For the current $r_2^{(t)}$, solve bank 1's FOC (2) for $\sigma^*(h_k; r_2^{(t)})$ at each grid point $h_k$, using a scalar root-finder (Brent's method on $\sigma - \frac{1}{1-h} - \frac{1}{\alpha(1-s(\sigma, r_2))} = 0$).

3. **Update $r_2$.** Compute shares $s_k = s(\sigma^*(h_k; r_2^{(t)}), r_2^{(t)})$ and update:

$$r_2^{(t+1)} = \frac{1}{1-\tilde{h}} + \frac{1}{\alpha} \cdot \frac{\sum_k (1-s_k)(1-h_k)\, p_k}{\sum_k s_k(1-s_k)(1-h_k)\, p_k}$$

where $\tilde{h} = \frac{\sum_k h_k\, s_k(1-s_k)\, p_k}{\sum_k s_k(1-s_k)\, p_k}$ and $p_k = P(h = h_k \mid x_j)$.

4. **Iterate** with damping ($r_2 \leftarrow 0.7\, r_2^{(t)} + 0.3\, r_2^{(t+1)}$) until $|r_2^{(t+1)} - r_2^{(t)}| < \varepsilon$.

*Remark* 11 (Computational structure). The inner solve (Step 2) requires $N_h$ scalar root-finding problems per iteration. Since $\sigma^*$ is monotone in $h$ for fixed $r_2$, each is well-conditioned. The outer loop (Steps 3–4) typically converges in 10–30 iterations with damping. The total cost is $N_x$ equilibrium solves, each requiring $\mathcal{O}(\text{outer iterations} \times N_h)$ root-finding calls. With the baseline parameters ($N_x = 5$, $N_h = 30$), the computation takes a few seconds.

## 4.3 Data Generation

Given the equilibrium, generate $N$ borrowers:

1. Draw $(x_i, h_i)$ from the joint distribution.
2. Look up equilibrium rates: $r_{1i} = \sigma^*(h_i; r_2^*(x_i))$ and $r_{2i} = r_2^*(x_i)$.
3. Draw winner: $W_i = 1$ with probability $s(r_{1i}, r_{2i})$, else $W_i = 2$.
4. Draw default: $D_i \sim \text{Bernoulli}(h_i)$.
5. Record observables: $(r_i^{\text{win}}, W_i, D_i, x_i)$, where $r_i^{\text{win}} = r_{1i}$ if $W_i = 1$ and $r_i^{\text{win}} = r_{2i}$ if $W_i = 2$.

## 4.4 Identification Checks

The simulated data is used to verify the key predictions from the identification argument:

1. **Bank 2 rate is constant within $x$.** Among bank 2 winners, all borrowers with the same $x$ should receive the same winning rate $r_i^{\text{win}} = r_2^*(x_i)$. This is the model prediction from Step 1 of the identification argument.
2. **Type recovery.** Among bank 1 winners, $E[D_i \mid r_i^{\text{win}}, x_i]$ should equal $h_i = \tau(r_i^{\text{win}}; r_2^*(x_i))$. In finite samples, the conditional default rate at each winning rate should track the underlying default probability.
3. **Linear identifying restriction.** The relationship

$$\ln\big(\alpha\, m(h; \mathbf{x}) - 1\big) = \lambda - \alpha\, \Delta(h; \mathbf{x}) \tag{14}$$

should hold across all $(h, x)$ pairs, where $m = \sigma^*(h) - \frac{1}{1-h}$ is the markup and $\Delta = \sigma^*(h) - r_2^*$ is the rate gap. In the $(\Delta, \ln(\alpha m - 1))$ plane, all points should lie on a line with slope $-\alpha$ and intercept $\lambda$.

4. **Adverse selection.** Bank 2 winners should have strictly higher default rates than bank 1 winners within every $x$-stratum: $E[D_i \mid W_i{=}2, x_i] > E[D_i \mid W_i{=}1, x_i]$.

The simulation code is implemented in `Code/model6/simulate_model6.m`.

## 4.5 Extensions

Several model extensions could improve estimation:

1. **Multi-product data.** If the same borrower has multiple loans (or a loan and a credit card), the model can be estimated across products. The same $(\mathbf{x}, \mathbf{x}')$ generates correlated rates across products, providing additional moments for identification.

2. **Panel data on the same borrower.** If the borrower's $\mathbf{x}$ changes over time (e.g., income changes, credit score updates) but their $\mathbf{x}'$ is stable (persistent account behavior), within-borrower variation in $r_2^*$ (driven by $\mathbf{x}$ changes) can be used to condition on $r_2$ over time.